IKP: PQC Migration

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Rule	Chatham House Rule. Information may be used without revealing the identity or affiliation of speakers or other participants, directly or indirectly.

This is a session record, not a transcript. Substantive discussion is retained with uncertainty and disagreement made explicit. Reported claims are not verified findings. Candidate agreements require working-group confirmation before use as decisions. Actions omit personal owners and identifying roles.

Agreements

The following sentences stand on their own. These are candidate agreements and reported points of convergence for working-group review. Inclusion does not establish consensus, verify a technical claim, or record formal adoption. The working group must confirm each decision before it is cited as such. Claims that were only discussed, including figures and alternatives that were not adopted, remain in full under Discussion.

BGIN convenes the competition as a neutral body. The target is methods of crypto agility for existing lattice and hash-based signatures, then a report, then a standard aimed at ISO.

New hardness assumptions are out of scope. A decade-scale standardization process is what those assumptions require.

The scoring baseline is ECDSA as deployed today. The measured object is the gap when post-quantum signatures and their proofs run on test nodes.

Evaluation criteria and a line-in-the-sand notion of "post-quantum ready in the present context" are the committee's job.

State is acceptable only where a monotonic counter is actually maintained, with a stateless fallback everywhere else. Within an epoch, repeating one message under a one-time key is safe; a second message is the danger; embedding the epoch in the message is one safety mechanism.

Hash-based implementations should be built from audited blocks (hypertree, leaves), not as a free-for-all.

Primitive swap is not migration. Readiness is system-level: keys, addresses, validation, legacy state, wallets, third parties, hardware, deployment, UX, and governance.

The five questions are the working playbook.

Key management after migration is a separate work item and belongs beside PQC migration in the upcoming workshops.

Choices made at migration time stick. Environmental cost belongs in the criteria.

Major chains, on the evidence offered, are not planning hybrid signatures; hybrid key establishment is a different recommendation.

The competition's participant track and its evaluation committee are different populations.

Documents considered

The session referred to the following documents, repositories, and standards. Bibliographic identifications and factual checks added during review are marked as editorial clarifications; they do not imply that the links were spoken or that the meeting adopted the findings. Names of the people who introduced the material are omitted.

Editorial terminology: WEIS is the Workshop on the Economics of Information Security, whose archive includes the 2002 meeting.

SLH-DSA and SPHINCS; ML-DSA; FN-DSA / Falcon; ECDSA as the baseline; Ed25519 (64-byte signature) and ML-DSA-44 (2,420-byte signature) as the visual comparison.

XMSS, WOTS, and one-time keys inside an epoch, as validator mechanics.

NIST standardization and parameter review, as the bar a retuned scheme would still have to clear. No FIPS number was spoken in this slice.

CRYPTREC, as the Japanese maintenance program.

ISO, as the intended destination of the eventual standard.

PQCVisualizer / Post-Quantum Signature Size Explorer: `https://pqcvisualizer.com/` URL identified editorially from the signature-size comparison.

BGIN GitHub repository intended to carry the five areas. The caption refers to a repository, not an additional document title.

GDC meeting report, location described as the BGIN website. No URL spoken.

Google Quantum AI and collaborators: Securing Elliptic Curve Cryptocurrencies against Quantum Vulnerabilities: Resource Estimates and Mitigations, with associated zero-knowledge proof material: `https://arxiv.org/abs/2603.28846` Separately, ECDSA.fail is an Eigen Labs project for optimizing quantum elliptic-curve circuit resource costs: `https://ecdsa.fail/` The benchmark is not the author of the Google paper or its proof, and neither a resource estimate nor a benchmark improvement demonstrates a practical compromise of deployed ECDSA. These are editorial identifications of public projects, not attributions of meeting remarks.

Science 2006 tutorial on information-security economics; an open-access workshop series from 2002, name garbled.

SHA-1 migration, as the only historical comparison point mentioned.

The git repository, the shared crypto-agility note, the SSR conference URL, and the HAWK candidate-count correction were not part of this slice's discussion. The LAMPS drafts were not named. No poster was recorded.

Discussion

The following is the substance of the session. It preserves claims, figures, alternatives, and objections. Discussion does not establish a decision. Any candidate agreement requires explicit working-group confirmation.

Competition scope. BGIN was announced as the neutral body that will help run a PQC-migration / crypto-agility competition, supported by NEDO, the Japanese agency identified in the published programme and competition briefing (an editorial correction to the caption). The competition is not a call for new signature schemes or new hardness assumptions. The schemes in scope are the ones already proposed, in two families: lattices and hash-based signatures. The work is to experiment with those schemes and to show crypto-agility methods for moving signatures on public permissionless blockchains. Duration stated: the next three years, or two and a half years. Agility is ongoing; the competition is meant to make the methods repeatable.

Intended output: a report, testing, and then a proposed standard, so that later agility has a defined method rather than a one-off replacement of "the brick at the bottom of the stack." Cryptography should move up the stack. Four challenges are what the prize will measure. They were not listed. An evaluation committee is being formed. It remains open to post-quantum cryptographers and to people who build standards. The committee is to write the criteria for "good crypto agility in a post-quantum world." The finish line, as stated: a line in the sand — after this process, in the current context, a system can be called post-quantum ready — and then an ISO standard, using standards relationships BGIN already has.

Timeline stated in the session. Earlier this year, at GDC: a high-level presentation from Google Quantum AI. A meeting report was said to exist and to be available via the BGIN website.

The previous weekend, at a foundation quantum retreat: strong technical ideas, and a concentration on hash-based signatures. The gap named: other schemes still need a wider forum. BGIN was proposed as that forum.

November: a call for operators of testbeds on public permissionless networks. An Ethereum testnet for hash-based signatures is to be folded in. Educational institutions connected to BGIN would host some of those nodes, so the evaluation has data for a report to the Japanese sponsor and to a second body whose acronym was garbled. A winner was projected in about two years.

December: an academic security-research conference, with a workshop in Baltimore. The call for papers is open. Four or five papers were already in hand. The workshop is to be co-chaired with the competition. Those papers become core material for the new year.

After that workshop: applications open for competition participants, as distinct from the evaluation committee.

2027: a six-to-eight-month period to define the competition while participants share agility methods.

2028: the award, after assessment of test configurations on blockchain testnets.

What is measured. The lowest-hanging measurement is signatures and public-key cryptography. The baseline is today's ECDSA-based systems. Post-quantum signatures, and proofs that they are in use, run on the new nodes. The gap between baseline and post-quantum operation is the dataset for the review, for naming a winning method, and for moving that method into a standard.

Scoring, and whether "no new schemes" survives contact with auto-research. A question: if witness size, verification cost, and similar figures are already known for SLH-DSA and ML-DSA, is the competition inventing schemes to optimize those figures? The first answer was no: known scores go into the review; new schemes are out of scope. That answer was then qualified. At the retreat, frontier auto-research models had already shrunk SPHINCS from a stated baseline of about 7,000 or 8,000 (the unit was spoken as bits) to about 5,000. The qualification: if blockchain-specific improvements appear, the criteria may have to admit them, or the competition's result will be irrelevant to chains that ship their own signatures. A counter-position: inventing or retuning a signature is a different activity from evaluating a fixed scheme or a protocol, and the group may not be ready. The clarification offered: not a new primitive; improvements to the chosen schemes. A further counter-position: NIST parameter changes are made by many cryptographers and need enough public cryptanalysis before anyone should be confident. Parameters that have not had that review should not be treated as ready. This was explicitly parked as an open question for the evaluation committee.

A standards-process perspective, kept here as subject matter rather than as attribution: standardized schemes can stay signatures, and trouble starts when zero-knowledge proofs or other layers are optimized on top. Some optimizations introduce vulnerabilities; some do not. "Score" was still unclear. Hybrid signatures are a live design space: two signatures side by side, one signature inside another, or intermediate optimizations. Opening the competition to those compositions requires many reviewers, not only a small evaluation committee.

Stateful signatures. The question was whether stateful schemes are in scope, given the risk of losing the counter and the hard limit on how many signatures a state may issue.

Validator consensus on Ethereum was described as almost certain to use stateful signatures, and as relatively safe. Validator networks have a global epoch clock that defines state per validator. State reuse is mitigated, not eliminated. State on the execution layer, in user wallets, was rejected: a backup (cloud backup was the example) desynchronizes the counter. Aggressive size optimization is why validators want state. A wallet client tracking state was called infeasible.

A Bitcoin-oriented hybrid of SPHINCS: use the stateful path only when state is certainly tracked — a key born inside a hardware wallet that does not export stateful signing, and that wallet keeps the counter. Every other path, including recovery from that wallet, uses the stateless scheme. State is confined to the place that can track it.

Institutions that already pay high fees and run redundant HSMs were expected, in this discussion, to prefer the stateless path and to dislike a monotonic counter in a cloud architecture. A holder

moving very large value would use stateless. Both paths were still wanted, because small payments and non-institutional users need the smaller stateful signatures if Bitcoin is to remain usable for everyday amounts. An illustrative ceiling of "10 million dollars" was immediately withdrawn as arbitrary and probably far too high: even stateless SLH-DSA may price out much smaller transfers. An institution that batches many of its own transactions might later accept stateful signatures because its own fees evict its own transactions. That outcome was left uncertain.

Address reuse differs by chain. Ethereum reuses keys more. Bitcoin is built to reuse less, and often reuses anyway. Because fee pressure will make people use state if state exists, the safer path was argued to be guidance for careful use, not a ban that will be ignored.

New hardness assumptions were set aside: they need many independent reviewers and on the order of a decade of standardization. Inside hash-based schemes, parameters can move further while the security argument stays intelligible, and recent work has been less conservative, including SLH-DSA with a smaller signature budget. A new hash-based scheme had been announced the day before the session; the claim was that a useful amount of function can be kept at decent security. Novel constructions should not ship straight to production. SLH-DSA was described as conservative in both parameters and design, with room left to optimize. Had the 2016 design of hash-based signatures had today's auto-research tools, it would look different. Lattice parameters should not be tuned as freely. ML-DSA was called fine. FN-DSA was called fine and also unsettling. Where fee cost dominates, hash-based schemes were preferred; SLH-DSA was preferred where cost does not dominate.

A preference was stated for a small number of reasonable hash-based options, with enough time for review, rather than a wide menu of primitives.

On implementation: hash-based schemes are assemblies of standard blocks. Choosing order and parameters can sit one layer above a formally verified, heavily audited implementation of a hypertree and of the leaves. That is not an invitation for each deployer to invent a scheme. It is a claim that the hash assumptions are strong enough to allow a limited composition layer.

Backup is the main operational failure of stateful hash-based signatures. Anyone who chooses state has to know how the counter is backed up and synchronized. Many deployments do need that synchronization.

Decision tree offered: the most conservative assumption combined with the smallest signatures is exactly the choice that requires state. The next question is whether a monotonic counter survives redundancy and backup. If the answer is anything but yes, do not use state. Use a larger lattice signature if size is the constraint, or a stateless hash-based signature if size can be spent. Large institutions are built so that an entire cloud region can fail without stopping operations. That architecture fights state. State-management tools exist. Randomness and state were called the two things cryptographic implementations handle badly.

Ethereum's reasonable case for state: consensus validators are already stateful. They must know whether they have voted, and they can be slashed for a bad vote. An epoch already runs across the network. Adding stateful cryptography matches an existing economic reason to keep state, even though

slashing is uncommon. Validators sign far more often than an execution-layer user (the example was a few signatures a week). They optimize hand size; the protocol optimizes conservatism; the combination named was stateful hash-based signatures.

Safety detail for one-time keys: inside one epoch, signing the same message repeatedly with a one-time key does not reveal the private key. A WOTS key, the base of an XMSS tree, can sign one message many times (a thousand was the figure) and publish the pairs safely. The break is a second, different message. In the validator design described, the epoch time is inside the message, so a signature from a later epoch is not valid for the earlier epoch. Leaf keys were described as no longer valid at that second epoch. That is a built-in redundancy, not a general license to reuse state.

NIST-style state guidance was described as strict, and as appropriately strict: effectively a single HSM. Relaxing it was not treated as a lobbying priority. The deployments most likely to step outside that guidance are blockchains, and they were expected to proceed whether or not a standards body agrees.

Migration as a system and as an incentive problem. A second block of the session separated primitive migration from system migration, and technical migration from application and business migration. Even if many quantum-safe primitives exist, doing all layers at once was called intractable. Stakeholders will not move on the same day. Incentive analysis is required to see what goes wrong when they move at different times.

Information-security economics was offered as a tool already about a quarter-century old, with a tutorial in *Science* in 2006 and an open-access workshop series started in 2002 (the series name is garbled). Two opposite research patterns have to be kept apart:

People fail to follow the economically optimal strategy, and the strategy is counterintuitive. The work corrects the intuition.

People follow the economically optimal strategy, and that strategy damages security. The work then has to recommend a social mechanism.

Theory models need empirical checks before they should drive a migration plan. Both patterns may be useful; using the wrong one would mis-state the problem.

CRYPTREC was described as the Japanese government's maintenance body for cryptographic standards. It does not simply import global standards; it re-analyzes them for domestic government use and publishes on a yearly cycle. Its present PQC coverage was described as example-based, not an exhaustive survey. Migration of particular application protocols, including blockchain protocols, was placed at the next step or the step after that — not as current CRYPTREC work. NIST competitions matter, and classical algorithms still need maintenance after they are standardized.

Two workshops were distinguished. The December Baltimore workshop was already on the program. A Japanese domestic workshop at the end of November was described as a problem-formulation meeting, not a solutions showcase. Position papers were requested. The technical history offered as a reason to formulate problems carefully:

Public timestamping and surety-style archives publish a root hash into a verifiable archive. The hash is a tiny artifact. Time resolution of a day or half a day, plus expensive verification at libraries, made that

design unusable as a currency rail. A later proposal used digital-television archives as the publication medium and ran into the same resolution and cost problems.

A blockchain uses cryptography in at least two or three places: a hash function; a signature on each transaction, flooded on the peer-to-peer network; and cryptography on the network path itself, including TLS at user interfaces. Some of those layers are not owned by the blockchain community. Migration therefore needs other technical communities in the room.

If the application is financial, the financial sector has its own calendar. The calendars have to be harmonized.

A more expensive stack can be a more energy-intensive stack. Migration was framed as a chance to revisit that cost, including a "greener" outcome, and also as a moment when a bad choice becomes sticky: after the migration, a major change is hard. Environmental load had not been on the competition's list and was asked to be added. "Graveyard" chains were mentioned as systems for which migration is almost a reset.

Advanced key management was named as the second topic of the workshop pair, because the key-management structure after migration is a different structure, not a leftover of the old one. That pairing was treated as aligned with custody-side key-management work.

System-level readiness and a five-question playbook. A comparison tool (a PQC visualizer; the domain as heard is unverified) was used to put a classical signature beside a post-quantum one: Ed25519 at 64 bytes, against ML-DSA-44 at 2,420 bytes. The objects are different enough to change transaction formats, storage, fees, and the infrastructure that verifies transactions. Adopting the scheme is not a replacement.

The contribution's claim: post-quantum migration is more than replacing signatures. A primitive sits in a system of keys, addresses, validation rules, wallets, third parties, hardware, and other infrastructure. Five dependency areas, also intended for a BGIN write-up:

1. Key and address lifecycle.
2. Transaction and validation rules.
3. Legacy state.
4. Ecosystem compatibility.
5. Deployment path.

A chain can adopt a post-quantum scheme and still have no operational migration path. Readiness is a system property. Five evaluation questions were proposed, and the room was asked whether a playbook could stay at five:

1. What cryptographic assumption is changing? This forces an inventory of primitives, keys, and the threat model.

2. What else depends on it? Third parties have to move. In production this is a UX and governance problem, including wallet infrastructure and operational dependencies. Hardware is often omitted.

3. Can old and new mechanisms coexist? Hybrid versus other transition styles. Early, and system-specific. The question was left open on purpose.

4. Who has to upgrade? Users, wallets, nodes, hardware, services, and other ecosystem participants.

5. Does the design make the next cryptographic transition easier, or does it only swap one frozen primitive for another? Existing accounts, dormant funds, and how much of the choice falls on the user. Chains are already hard to use; the transition should not dump an unbounded menu on the user.

Availability of a primitive and readiness to operate it were separated. The questions to add are whether the cryptography can be changed safely and whether the system stays adaptable.

There is no single migration benchmark, because the evaluation is per system. Interoperability was proposed as a risk that might already sit inside "what else depends on it," and that should not be lost: it is a core property of these networks. Hybrid signatures drew a direct empirical claim: very little appetite on the major chains; no chain the discussion had encountered is planning hybrid signatures; adding a hybrid layer and later removing it is two migrations. Conservative hash-based choices were the reason an elliptic-curve layer was felt unnecessary. Key exchange was explicitly different: hybrid key exchange was called valuable. A request for the shape of NIST's hybrid recommendation was deferred to the next session. Constrained wallets and constrained hardware, often built around elliptic-curve point operations, were expected to struggle. A device mentioned as newly available was not identified.

On question 5, the only earlier migration available as a dataset was the move away from SHA-1 and similar hashes, and that was a long time ago. This migration was expected to be repeated at least through the post-quantum period, so the data gathered now is what would show whether a later migration is easier. A counter-point: systems are more complex now than at the last migration, so "easier next time" may be false, and the question may really be about complexity and about mutual dependencies among uses of cryptography. "Easier" was split in two: the underlying cryptography is easier because it has been done once, or the method is easier because this competition standardized the method. Breaking the question along dependencies was accepted as useful. The five-question list was kept as the working playbook unless a chain-specific difference forces an addition.

The session then broke for about ten minutes, with a restart announced in five, on the same remote link.

Matters not resolved

Are blockchain-specific parameter changes and auto-research shrinks of SPHINCS inside a competition that set out to forbid new schemes? Parked for the evaluation committee. Confidence requires cryptanalytic review; AI-shrunk parameters do not arrive pre-trusted.

The "four challenges" the prize measures were never enumerated.

Stateful signatures: in scope for validators and for hardware that truly owns the counter; out of scope for ordinary wallets and, on the argument in the room, uncomfortable for multi-region institutional HSMs. Whether institutions will ever pay for the stateful path to avoid fee self-competition was left open.

Whether to lobby for looser state-management rules. Current single-HSM guidance was defended. A lobbying campaign was not endorsed.

New hardness assumptions: out. How far hash-based parameters may move, versus lattice parameters: hash-based has more room; lattices should stay put. FN-DSA's floating-point behavior remains a concern.

Hybrid signatures on major chains: the claim in the room was that appetite is minimal and that a hybrid layer creates a second migration. Key-exchange hybrid was not lumped into that claim. Coexistence of old and new mechanisms was left as an open evaluation question.

Whether five playbook questions are enough. Interoperability may need to be explicit. "Easier next time" needs a definition: cryptographic complexity, or a standardized method. System complexity may make the next migration harder.

When each stakeholder moves, and which economic story applies (people ignore the optimal strategy, or people follow it and harm security). Not resolved.

CRYPTREC is not, on this account, doing blockchain protocol migration now. Whether and when it will was left as a later step.

Energy cost of a heavier post-quantum stack, and whether migration should be used to retire other accumulated design problems. Raised, not specified.

What a migration playbook document literally contains, beyond the five questions. No single template was agreed.

Disposition of the work

Neutral convener: BGIN. Sponsor: a Japanese government agency, name unverified in the audio. A second recipient of the eventual report was named only by a garbled acronym.

Committee: open. Some members already participating.

Schemes: lattices and hash-based signatures. Ethereum hash-based-signature testnet work is to be folded into institution-hosted nodes.

Known scores for SLH-DSA and ML-DSA will be reused. SPHINCS size was reported as already reduced by auto-research from roughly 7,000–8,000 to roughly 5,000 in the unit spoken.

A new hash-based scheme was said to have been announced the day before. No stable name was recoverable.

CRYPTREC: yearly maintenance, example-based PQC coverage, protocol migration not current work.

Playbook: five areas and five questions, headed for a BGIN document. No benchmark number exists yet.

GDC meeting report: said to be on the BGIN website.

Actions

Each action is written so that it can be tracked without the rest of the record. Where the room named only a person, the action is kept and the name is omitted.

November: call for permissionless-network testbeds; fold in the Ethereum hash-based testnet; institutions host nodes.

End of November: Japanese domestic workshop aimed at problem statements, not solution talks.

December: Baltimore workshop at the security-research conference; four or five papers already in; workshop them, then open participant applications.

2027: six to eight months defining the competition and collecting agility methods.

2028: assess testnets and make the award; move the winning method toward a standard and ISO.

Evaluation committee: decide whether blockchain-specific parameter changes are in scope; require broad review of any composition or hybrid; write state guidance that distinguishes validators and capable hardware from wallets; keep new hardness assumptions out; consider adding energy, interoperability, and key management; try to keep the playbook at five questions, and define "easier.".

Gather migration data now so a later transition can be compared. The SHA-1 transition is too old to serve as that dataset alone.

Coverage

This record covers the published first PQC migration session in Room A. An earlier working timetable had placed the long PQC sessions in Room B. The published programme and this recording both place them in Room A. There is no Day 1 Room B recording, and the published timetable leaves Room B empty on Day 1. Figures are reported session claims, not independently validated measurements. Technical names are corrected where supported by the recording, chat, or identified primary documentation. Where a name could not be checked against another part of the same recording, it is left unverified rather than guessed. Logistics, introductions, and names of speakers are omitted.

Audio checks, a tea break (ten minutes, then "back in five"), screen-share setup, thanks, and long stretches of Japanese that the caption track did not translate. Where a later English restatement recovered a technical point (validator epochs, one-time keys, WOTS, slashing), that point is included above. Untranslated Japanese is not guessed into extra claims. Lunch was not in this slice.

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Source clarifications and open checks

The Japanese competition agency is identified editorially as NEDO, using the published programme and competition briefing.

"JBCA" or similar, as a co-recipient of the evaluation report.

Unit "bits" on the 7,000–8,000 to ~5,000 SPHINCS shrink. The numbers are reported as spoken.